VELS
INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR

GrowthAI: A skill-gap driven personal growth and time-management assistant with rule-based adaptive planning and advisory task-completion prediction

ABSTRACT

Setting a career goal is easy, but being able to identify the skills gap and study the missing skills on every other day is more challenging. This paper presents GrowthAI, a web application that joins them into a whole, instead of treating them as separate processes. A learner sets a goal and the system recommends a roadmap based upon the skills they need to develop. A timed multiple-choice assessment with server-controlled time, shuffled questions and options, and limited attempts measures the current level. Thresholds of fixed scores distinguish beginner, intermediate and advanced levels of skills, and the lower their present level, the higher they appear on the roadmap. Tasks are suggested daily based on available time, recent energy check-ins and skipping history, and every skipped task is recorded with its reason. Both the planner and difficulty recommendation follow explicit rules. A separate logistic-regression module provides an advisory estimate of whether a pending task will be completed. It does not interfere with the plan, and it only returns a value when the task log contains enough completed and skipped examples. The prototype has used React, FastAPI and PostgreSQL, together with bcrypt password hashing, signed JWT tokens and ownership checks. A functional test with the Data Analyst goal showed that the modules work together, but no user study has been performed, and an LLM-based coach and a weekly reflection remain future work.

Key words: Personal growth assistant, Skill-gap analysis, Adaptive task planning, Logistic regression, Web application